

NATIONAL OCCUPATIONAL STANDARD FOR WOOD PROCESSING TECHNOLOGIST



NOS.WPT.01
FIRST EDITION

APPROVING AUTHORITY

This National Occupational Standard (NOS) has been prepared and published under the authority of the Zambia Qualifications Authority Board on 15th February 2024.

ZAMBIA QUALIFICATIONS AUTHORITY

The Zambia Qualifications Authority (ZAQA) Act No. 13 of 2011 was enacted by the Government of the Republic of Zambia to ***“provide for the development and implementation of a national qualifications framework; establish the Zambia Qualifications Authority; provide for the registration and accreditation of qualifications; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing”***. Among other functions, ZAQA is responsible for ***determining national standards for any occupation***, through the various sector specific National Occupational Standards Development Teams (NOSDTs).

REVISION OF NATIONAL OCCUPATIONAL STANDARDS

National Occupational Standards (NOS) shall be revised every 5 years, or whenever it is deemed necessary, by the issuance of either amendments or of revised editions. It is important that users of the NOS ascertain that they are in possession of the latest amendments or editions.

NOS DEVELOPMENT TEAM RESPONSIBLE

This National Occupational Standard was prepared by the Manufacturing NOSDT, upon which the following organisations were represented:

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2. Copperbelt University (CBU)
3. Engineering Institution of Zambia (EIZ)
4. Good Time Steel Company Zambia Limited (GTS)
5. Lusaka Vocational and Technical College (LVTC)
6. Makeni Ecumenical Centre (MEC)
7. Ministry of Commerce, Trade and Industry (MCTI)
8. Ministry of Health (MoH)
9. National Institute for Scientific and Industrial Research (NISIR)
10. Northern Technical College (NORTEC)
11. University of Lusaka (UNILUS)
12. University of Zambia (UNZA)
13. Zambia Association of Manufacturers (ZAM)
14. Zambia Forestry College (ZFC)
15. Zambia Qualifications Authority (ZAQA) – Secretariat

ACKNOWLEDGEMENT

The Zambia Qualifications Authority would like to acknowledge the invaluable support of the following stakeholders that participated in the development of this NOS:

1. Mrs. Nancy Muyanje Malwa Mushota (Zambia Association of Manufacturers)
2. Ms. Hilary Moono Siamuzyulu Chibiya (Ministry of Health)
3. Mr. Bonaventure Gwelengwe (Biomedical Society of Zambia)
4. Mr. Machiko Alfred (Biomedical Society of Zambia)
5. Mr. Kelly Mwayengo (University of Lusaka)
6. Mr. Patrice Shawa (Zambia Forestry College)
7. Mr. Pascal Sikainga (Makeni Ecumenical Centre)
8. Mr Moses Ngosa (Ministry of Commerce, Trade and Industry)
9. Eng. Mark Kapasa Mutimushi (Northern Technical College)
10. Dr. James Phiri (University of Zambia)
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FOREWORD

The Zambia Qualifications Authority (ZAQA) is a statutory body under the Ministry of Education established by ZAQA Act No. 13 of 2011 to ***“develop and implement a national qualifications framework; register and accredit qualifications; and ensure that standards and registered qualifications are internationally comparable.”***

Among other functions, ZAQA is responsible for ***“determining national standards for any occupation”***, through the various sector specific National Occupational Standards Development Teams (NOSDTs) with experts composed of representation from the appropriate authorities, government departments, consumer associations, regulators, industry, academia, and non-governmental organisations, etc.

This National Occupational Standard (NOS) has been developed by the Manufacturing National Occupational Standards Development Team in accordance with the laid down procedures and guidelines of ZAQA. All users should ensure that they have the latest edition of this publication as the National Occupational Standards are revised from time to time.

This NOS shall be used by, among others, industry, employers, quality assurance bodies, awarding and professional bodies, and education and training institutions, as a benchmark to identify training needs, develop job profiles/descriptions, develop curricula, and learning programmes in various sectors where the occupation exists. In the Manufacturing sector, demonstration of competence against this NOS may be required in order to run a business or practice a craft or profession.

JUSTIFICATION

A Wood Processing Technologist is involved in converting logs and other raw wood materials into timber, veneer, plywood, block board, and a wide variety of finished wood products. Wood Processing Technologists are responsible for planning operations, designing the products to be produced, acquiring the raw materials, and servicing the woodworking machines and maintaining tools. This National Occupational Standard highlights the core knowledge, skills, competences and personal attributes that Wood Processing Technologist must possess to be successful in their jobs.

ACRONYMS AND ABBREVIATIONS

CS	Core Skills
K	Knowledge and Understanding
NOS	National Occupational Standard
NOSDT	National Occupational Standards Development Team
OK	Organisational Context
OSHE	Occupational Safety, Health and Environment
PC	Performance Criteria
PS	Professional Skills
RK	Regulatory Context
RPL	Recognition of Prior Learning
S	Skills
TK	Technical Knowledge
WPT	Wood Processing Technologist
ZAQA	Zambia Qualifications Authority
ZQF	Zambia Qualifications Framework

GLOSSARY OF TERMS

For the purposes of this NOS, the following terms and definitions shall apply:

Core Skills/Generic Skills: are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the NOS, these include communication related skills that are applicable to most job roles.

Disability: is the physical or mental impairment that substantially limits one or more major life activities.

Function: is an activity necessary for achieving the key purpose of the sector, occupation or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of the NOS.

Job Title: defines a unique set of functions that together form a unique employment opportunity in an organisation.

Knowledge and Understanding: are statements which together specify the technical, generic, professional, and organisational specific knowledge that an individual needs in order to perform to the required standard.

National Occupational Standards (NOS): are statements of the standards of performance individuals must achieve when carrying out functions in the workplace, together with specifications of the underpinning knowledge and understanding. They are precise descriptions of what an individual is expected to be able to do in his/her work role.

National Occupational Standards (NOS) Code: is a unique reference code that identifies a NOS.

National Occupational Standards Development Team (NOSDT): means an established group of national stakeholders/experts responsible for the development of National Occupational Standards within a specific economic sector or occupation.

Occupation: is a set of job roles which perform similar/related set of functions in an industry.

Organisational Context: includes the way the organisation is structured and how it operates, including the extent of operative knowledge that managers have in their relevant areas of responsibility.

Performance Criteria: are statements that together specify the standard of performance required when carrying out a task.

Scope: is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.

Sector: is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy, whose components share similar characteristics and interests.

Sub Sector: is derived from a further breakdown based on the characteristics and interests of its components.

Technical Knowledge: is the specific knowledge needed to accomplish specific designated responsibilities.

Unit Title: gives a clear overall statement about what the incumbent should be able to do.

Elements: set out competences the incumbent should possess to carry out the day-to-day activities.

1. OVERVIEW

This is an introductory section providing a summary and specific information or commentary about the content of the NOS and the targeted sector and occupation to help the user judge whether it is relevant to them.

NOS Code	NOS. WPT.01
Occupation	Wood Processing
Job Title	Wood Processing Technologist
Job Description	A Wood Processing Technologist is a person responsible for planning and designing wood products to be produced, acquiring the raw materials, supervising workers, and servicing the woodworking machines. He/she also performs inspections on processed materials and finished wood products for quality.
Job Purpose	A Wood Processing Technologist is responsible for converting raw wood materials into timber, veneer, plywood, block board, and a wide variety of finished wood products.
ZQF Level	6
Sector	Manufacturing
Sub sectors	Wood Product Manufacturing
Other Economic Sector(s) in which the Occupation is Practiced	Mining, Energy, Construction, Education, Government Ministries and Agencies, etc.
Other Similar Jobs that can be Performed in the Occupation	Carpenter, Tutor/Trainer and consultancy.
Minimum Educational Job Entry Qualification(s)	Diploma
Practicing License Requirements (if any)	Membership with: 1. The Zambia Institute of Manufacturers 2. Engineering Institution of Zambia
Training/RPL	1. Use of ICTs (Internet, Computer packages, email, Computer Software and Hardware necessary for the job.) 2. How to sketch manually and transfer to CAD 3. Drawing Quality Enhancement Methods
Minimum Job Entry Age	21 years
Prior Experience (Suggested)	Minimum of 1 year internship in wood processing industries
Performance Criteria	As described in the Units under Section 4

2. SCOPE

This NOS highlights the core competences, skills, knowledge and understanding, and personal attributes that Wood Processing Technologists are required to possess to be successful in their jobs.

3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES)

This job requires an individual to possess ability to apply advanced mathematical principles and statistics to solve problems, exceptional technical and problem-solving skills, as well as reasoning ability. He/she must have the ability to communicate effectively and clearly, as well as plan and prioritise their work. The individual must be courteous, creative, self-motivated, a great team worker, quality conscious, Occupational Health and Safety orientated, and physically fit. He/she must have experience in using AutoCAD, and ability to use fingers, hands, and feet with ease to complete the assigned task (dexterity).

4. UNITS AND ELEMENTS

This National Occupational Standard is divided into 5 Units, representing the tasks that a jobholder should undertake in his/her day-to-day work. Each unit is further broken down into elements depicting the number of activities to be carried out for the successful execution of a particular task.

UNIT 1. This Unit covers the skills and knowledge required by a Wood Processing Technologist to work according to personal safety, health, and environmental rules and protocols at the working site]

Unit No.	01
Unit Title	Health, Safety, and Environmental Rules and Protocols
Description	This Unit describes the skills and knowledge required by a Wood Processing Technologist to work according to safety, health, and environmental rules and protocols at the site.
Scope	This Unit covers the following: • Following Safety and Health Rules and Regulations. • Operations Conformity to Environmental Protection Standards. • Safety of Employees, Plant, and Equipment.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Following Safety and Health Rules and Regulations	To be competent, the individual must be able to: PC1. Interpret and implement the safety and health policies of the company. PC2. Interpret and implement the safety and health regulations of the company. PC3. Identify hazards and risks of all aspect of the operations of the plant and equipment.
Operations Conformity to Environmental Protection Standards	To be competent, the individual must be able to: PC4. Implement the safety and health policies of the company. PC5. Implement the safety and health regulations of the company. PC6. Eliminate hazards and risks of all aspects of the operations of the plant and equipment. PC7. Observe environmental protection standards of the company.
Safety of Employees, Plant, and Equipment	To be competent, the individual must be able to: PC8. Apply the standard operating procedures for the employees, plant, and equipment of the company. PC9. Control hazards and assess risks of all aspect of the operations of the plant and equipment.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of the company's: OK1. Safety and health policies. OK2. Environmental policies and regulations. OK3. Standard operating procedures.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Environmental risk assessment. TK2. Safety and health risk assessment of the workstation. TK3. Plant and equipment operation risk assessment.

C. Regulatory Context (knowledge of rule and regulations)	The individual on the job must demonstrate knowledge and understanding of the: RK1. Factories Act and other regulations. RK2. Engineering Institution of Zambia. RK3. Mine Safety Act and other regulations. RK4. Occupational Health and Safety Act and regulations. RK5. Workers' Compensation Act and other regulations. RK6. Explosives Act and other regulations. RK7. Environmental Management Act. RK8. Employment Code Act.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English (at least working level) and/or have the means to give simple instructions in the local language used at the workshop. CS2. Prepare and provide clear and simple instructions, details, and sketches to co-workers.
	Reading Skills
	The individual on the job must be able to: CS3. Read English, and/or have the means to give simple instructions in the local language used at the workshop. CS4. Read and interpret sketches, drawings or instructions provided for the required work.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS5. Listen attentively and interpret communication/ instructions from the supervisor and co-workers. CS6. Convey information clearly and concisely to co-workers.
B. Professional Skills	Decision-Making
	The individual on the job must be able to: PS1. Carry out safety, health, and environmental risk assessments (in conjunction with the Safety department) PS2. Identify and solve health, safety, and environmental issues.
	Plan and Organise
	The individual on the job should be able to: PS3. Plan and organise safety, health, and environmental awareness meetings. PS4. Demonstrate the use of other safety and health tools and equipment.
	Customer Centricity
	The individual on the job should be able to: PS5. Follow the company's code of conduct policy. PS6. Manage relationships with customers with intent to satisfying their requirements.

	Problem-Solving
	The individual on the job should be able to:
	PS7. Recognise problems and search for solutions.
	PS8. Choose best methods to solve the problem.
	PS9. Approach relevant authorities when required.
	Analytical Thinking
	The individual on the job should be able to:
	PS10. Apply domain knowledge and observation to select the right course of action.
	Critical Thinking
	The individual on the job should be able to:
	PS11. Critically evaluate information of incidences obtained from supervisors, co-workers, and other workers during day-to-day operations.
	PS12. Ask relevant questions for better understanding from supervisors, co-workers, and other workers on the risks.

UNIT 2 [This Unit covers the skills and knowledge required by a Wood Processing Technologist to source for production materials]

Unit No.	02
Unit Title	Sourcing of Production Materials
Description	This Unit describes the skills and knowledge required by a Wood Processing Technologist to plan and acquire the right materials to process and produce required finished wood products.
Scope	This Unit covers the following: - Identifying the Correct Materials to be Used in the Factory. - Recommending the Right Materials to be Used in the Factory to Management. - Facilitating the Procurement Process of the Materials to be Used in the Factory.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Identifying the Correct Materials to be Used in the Factory	To be competent, the individual must be able to: PC1. Identify the correct type of raw wood materials. PC2. Determine the correct type of additives and binders. PC3. Recognise the wood strength properties. PC4. Specify the type of material to be used in the factory. PC5. Quantify the materials to be used in the production cycle.
Recommend the Right Materials to be Used in the Factory to Management	To be competent, the individual must be able to: PC6. Recommend the correct type of raw wood materials. PC7. Recommend the correct type of additives and binders. PC8. Specify the type of material to be used in the factory. PC9. Quantify the materials to be used in the production cycle.
Facilitating the Procurement Process of the Material to be Used in the Factory	To be competent, the individual must be able to: PC10. Provide the specifications of the materials to be used in the factory to management. PC11. Verify that the procured materials are the correct ones.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Various types of wood materials. OK2. Strength properties of different wood types. OK3. Specific types of materials to be used in the factory. OK4. Quantifying materials to be used in the production cycle. OK5. Strategic direction and objectives/targets for the company.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. How to identify the correct type of raw wood materials. TK2. Wood strength properties. TK3. Identifying the correct types of additives and binders to be used.

	TK4. Preparation of various types of material specification documents to management. TK5. Quantifying the materials to be used in the production cycle.
C. Regulatory Context (knowledge of rules and regulations)	The individual on the job must demonstrate knowledge and understanding of the: RK1. Factories Act and other regulations. RK2. Engineering Institution of Zambia. RK3. Mine Safety Act and other regulations. RK4. Occupational Health and Safety Act and regulations. RK5. Workers' Compensation Act and other regulations. RK6. Explosives Act and other regulations. RK7. Environmental Management Act. RK8. Employment Code Act.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English (at least working level) and/or have the means to give simple instructions in the local language used at the workshop. CS2. Prepare and provide clear and simple instructions, details and sketches to co-workers.
	Reading Skills
	The individual on the job must be able to: CS3. Read English, and/or have the means to give simple instructions in the local language used at the workshop. CS4. Read and interpret sketches, drawings or instructions provided for the required work.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS5. Listen attentively and interpret communication/ instructions from the supervisor and co-workers. CS6. Convey information clearly and concisely to co-workers.
B. Professional Skills	Decision-Making
	The individual on the job must be able to: PS1. Follow organisational rule-based decision-making processes. PS2. Take decision with systematic course of actions and/or responses.
	Plan and Organise
	The individual on the job should be able to: PS3. Plan and organise work to meet the set targets. PS4. Work and coordinate with others.
	Customer Centricity
	The individual on the job should be able to: PS5. Follow the company's code of conduct policy.

	PS6. Manage relationships with customers with intent to satisfying their requirements.
	Problem-Solving
	The individual on the job should be able to:
	PS7. Recognise problems and search for solutions.
	PS8. Choose best methods to complete assigned tasks.
	PS9. Approach relevant authorities when required.
	Analytical Thinking
	The individual on the job should be able to:
	PS10. Apply domain knowledge, observations, and select course of action to undertake.
	PS11. Apply risk-based approach to analyse situations for actions.
	Critical Thinking
	The individual on the job should be able to:
	PS12. Determine the implications of failure to conform to desired safety, health, and environmental regulations and impact on the employees and the company.

UNIT 3 [This Unit covers the skills and knowledge required by a Wood Processing Technologist to supervise and coordinate the working team and production process]

Unit No.	03
Unit Title	Supervision of Working Team
Description	This Unit describes the skills and knowledge required to by a Wood Processing Technologist to supervise and coordinate the working team and production process.
Scope	This Unit covers the following: • Supervising Workers. • Providing Oversight on the Different Components of the Production Process.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Supervising Workers	To be competent, the individual must be able to: PC1. Identify all the production lines. PC2. Observe and direct the execution of work as planned. PC3. Monitor and guide the workers in all the production lines.
Providing Oversight on the Different Components of the Production Process	To be competent, the individual must be able to: PC4. Demonstrate knowledge of all the components of the production process. PC5. Demonstrate understanding of activities involved in each component of the production process. PC6. Follow all controls in the production lines.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Production processes. OK2. Activities involved in each component of the production process. OK3. Company standard operating procedures.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of the: TK1. Different activities within their work area. TK2. Applicable techniques of work. TK3. Quality, timelines, safety, and risks at the site. TK4. Importance and need to support workers facing challenges in the workflow.
C. Regulatory Context (knowledge of rule and regulations)	The individual on the job must demonstrate knowledge and understanding of the: RK1. Factories Act and other regulations. RK2. Engineering Institution of Zambia Act. RK3. Mine Safety Act and other regulations. RK4. Occupational Health and Safety Act and regulations. RK5. Workers' Compensation Act and other regulations. RK6. Explosives Act and other regulations. RK7. Environmental Management Act. RK8. Employment Code Act.

Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to:
	CS1. Write in English (at least working level) and/or have the means to give simple instructions in the local language used at the site.
	CS2. Prepare and provide clear and simple instructions, details, and sketches to co-workers.
	Reading Skills
	The individual on the job must be able to:
B. Professional Skills	CS3. Read English, and/or have the means to give simple instructions in the local language used at the workshop.
	CS4. Read and interpret sketches, drawings or instructions provided for the required work.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to:
	CS5. Listen attentively and interpret communication/ instructions from the supervisor and co-workers.
	CS6. Convey information clearly and concisely to co-workers.
	Decision-Making
	The individual on the job must be able to:
	PS1. Supervise the workers.
	PS2. Monitor different components of the production process.
	Plan and Organise
	The individual on the job should be able to:
	PS3. Plan schedules and supervise the workers.
	PS4. Coordinate different components of the production process.
	Customer Centricity
	The individual on the job should be able to:
	PS5. Complete all assigned tasks in coordination with team members.
	Problem-Solving
	The individual on the job should be able to:
	PS6. Identify problems and search for solutions.
	PS7. Choose the best methods to solve the problem.
	PS8. Approach relevant authorities when required.
	Analytical Thinking
	The individual on the job should be able to:
	PS9. Apply domain knowledge and observation to select the right course of action.
	Critical Thinking
	The individual on the job should be able to:
	PS10. Critically evaluate information obtained from workers and co-workers involved in the operation process.

UNIT 4 [This Unit covers the skills and knowledge required by a Wood Processing Technologist in processing of wood products]

Unit No.	04
Unit Title	Production of Wood Products
Description	This Unit describes the skills and knowledge required by a Wood Processing Technologist to convert logs and other raw wood materials into timber, veneer, plywood, block board, and other finished wood products.
Scope	This Unit covers the following: • Conducting Pre-processing Wood Procedures. • Production of Wood into Required Finished Products.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Conducting Pre-Processing Wood Procedures	To be competent, the individual must be able to: PC1. Quantify the materials to be used in the production line. PC2. Identify the cutting patterns to be used. PC3. Interpret the cutting, edging, and planning techniques to be used. PC4. Assess the quality of the raw materials.
Production of Wood into Required Finished Products	To be competent, the individual must be able to: PC5. Identify the products to be manufactured. PC6. Interpret the designs of components to be produced. PC7. Carry out machine setting. PC8. Cut the wood into the required classifications. PC9. Assemble the wood into the required products (veneer boards, plywood, block board, cabinets, etc.). PC10. Carry out the finishing works (sanding, varnishing, and drying). PC11. Store the produced wood products correctly in the right place. PC12. Recycle some wood wastes into valuable products, such as chipboards, particle boards, etc. PC13. Service the machines after use.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of the company's: OK1. Standard operating procedures. OK2. Rules and practices. OK3. Vision, mission, and values. OK4. Strategic direction and objectives/targets.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Local manufacturing codes and mandatory standards for producing the required wood products. TK2. Preparing various types of job specification documents for job requirements. TK3. Plant and equipment operational risk assessments.

	TK4. Hazards associated with the operational activities. TK5. Accessing computer drawing using Auto-Cad software (used for viewing designs drawings). TK6. Sequencing of work schedules.
C. Regulatory Context (knowledge of rule and regulations)	The individual on the job must demonstrate knowledge and understanding of the: RK1. Factories Act and other regulations. RK2. Engineering Institution of Zambia Act. RK3. Mine Safety Act and other regulations. RK4. Occupational Health and Safety Act and regulations. RK5. Workers' Compensation Act and other regulations. RK6. Explosives Act and other regulations. RK7. Environmental Management Act. RK8. Employment Code Act.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English (at least working level) and/or have the means to give simple instructions in the local language used at the site. CS2. Prepare and provide clear and simple instructions, details and sketches to co-workers
	Reading Skills
	The individual on the job must be able to: CS3. Read English, and/or have the means to give simple instructions in the local language used at the workshop. CS4. Read and interpret sketches, drawings or instructions provided for the required work.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS5. Listen attentively and interpret communication/ instructions from the supervisor and co-workers. CS6. Convey information clearly and concisely to co-workers.
B. Professional Skills	Decision-Making
	The individual on the job must be able to: PS1. Follow the organisational rule-based decision-making processes. PS2. Take decision with systematic course of actions in consultation with the superiors.
	Plan and Organise
	The individual on the job should be able to: PS3. Plan and organise work to meet the set targets. PS4. Work and coordinate with others.
	Customer Centricity
	The individual on the job should be able to: PS5. Follow the company's code of conduct policy.

	PS6. Manage relationships with customers with intent to satisfying their requirements
	Problem-Solving
	The individual on the job should be able to:
	PS7. Identify problems and search for solutions.
	PS8. Choose best methods to complete assigned tasks.
	PS9. Approach relevant authorities when required.
	Analytical Thinking
	The individual on the job should be able to:
	PS10. Apply domain knowledge, observations, and data to select course of action to perform tasks.
	PS11. Apply risk-based approach to analyse situations for actions.
	Critical Thinking
	The individual on the job should be able to:
	PS12. Critically evaluate information obtained from customers, supervisor, and co-workers to perform day-to-day activities.
	PS13. Ask relevant questions for better understanding.

UNIT 5 [This Unit covers the skills and knowledge required by a Wood Processing Technologist to engage in wood processing entrepreneurship]

Unit No.	05
Unit Title	Entrepreneurship in Wood Processing
Description	This Unit describes the skills and knowledge required by Wood Processing Technologist that would enable him/her to become self-reliant, innovative, and create wealth and jobs.
Scope	This Unit covers the following: • Introducing Entrepreneurship. • Identifying Entrepreneurial Opportunities in the Environment. • Developing New Products. • Developing a Business Plan.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Introducing Entrepreneurship	To be competent, the individual must be able to: PC1. Identify the principles of entrepreneurship. PC2. Recognise the types of entrepreneurs. PC3. Explain the characteristics of entrepreneurs. PC4. Explain the role of entrepreneurship.
Identifying Entrepreneurial Opportunities in the Environment	To be competent, the individual must be able to: PC5. Explain entrepreneurship opportunity theories. PC6. Identify entrepreneurial opportunities in the environment. PC7. Recognise factors affecting entrepreneurship opportunities.
Developing New Products	To be competent, the individual must be able to: PC8. Apply new product development strategies. PC9. Apply new product development processes.
Developing a Business Plan	To be competent, the individual must be able to: PC10. Explain the features of a business plan. PC11. Develop a business plan. PC12. Implement a business plan accordingly.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Relevant legislation, standards, policies, and procedures relevant to work. OK2. Who to approach for support to obtain work related information, clarifications, and support. OK3. Documentation and related procedures applicable in the context of work.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Entrepreneurial skills. TK2. Use of IT statistical and analytical tools. TK3. Business standard operating procedures.

	TK4. Business planning. TK5. Product and market research.
C. Regulatory Context (knowledge of rule and regulations)	The individual on the job must demonstrate knowledge and understanding of the: RK1. Factories Act and other regulations. RK2. Engineering Institution of Zambia Act. RK3. Mine Safety Act and other regulations. RK4. Occupational Health and Safety Act and regulations. RK5. Workers' Compensation Act and other regulations. RK6. Explosives Act and other regulations. RK7. Environmental Management Act. RK8. Employment Code Act. RK9. Company Registration Act and other regulations.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English (at least working level) and/or have the means to give simple instructions in the local language used at the site CS2. Prepare and provide clear and simple instructions, details and sketches to co-workers
	Reading Skills
	The individual on the job must be able to: CS3. Read English, and/or have the means to give simple instructions in the local language used at the workshop. CS4. Read and interpret sketches, drawings or instructions provided for the required work.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS5. Listen attentively and interpret communication/ instructions from the supervisor and co-workers. CS6. Convey information clearly and concisely to co-workers.
B. Professional Skills	Decision-Making
	The individual on the job must be able to: PS1. Follow organisational rule-based decision-making processes. PS2. Take decision with systematic course of actions and/or responses.
	Plan and Organise
	The individual on the job should be able to: PS3. Plan and organise production and marketing strategic planning meetings. PS4. Organise work schedules to meet deadlines. PS5. Work constructively and collaboratively with others.

	Customer Centricity
	The individual on the job should be able to:
	PS6. Follow the company's code of conduct policy.
	PS7. Manage relationships with customers with intent to satisfying their requirements.
	Problem-Solving
	The individual on the job should be able to:
	PS8. Recognise problems and search for solutions.
	PS9. Choose best methods to solve the problem.
	PS10. Approach relevant authorities when required.
	Analytical Thinking
	The individual on the job should be able to:
	PS11. Apply domain knowledge and observation to select the right course of action.
	Critical Thinking
	The individual on the job should be able to:
	PS12. Critically evaluate information of incidences obtained from customers, supervisors, co-workers, and workers during day-to-day operations.
	PS13. Ask relevant questions for better understanding from supervisors, co-workers, and workers on the risks.

5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS

These include, but not limited to:

Equipment and Tools:

Design and prototyping equipment and tools; computer software applications; machine shop equipment and tools; fabrication shop equipment and tools; electrical equipment; maintenance equipment and tools; testing equipment and tools; personal protective equipment, etc.

Raw Materials and Consumables:

Wood, chemicals, additives, nails, lubricants, components, spare parts, glues, etc.

6. JOBHOLDER DILEMMAS/CHALLENGES AND COMPLEXITIES

Wood Processing Technologists face challenges such as: obsolete and/or inappropriate equipment and tools; budgetary constraints; inadequate product costing skills; poor technical skill; bureaucracy in procurement procedures; lack of appreciation of preventive maintenance by non-engineering management staff; labour intensive nature of the work; rapid changing of technology and materials; and lack of personal protective equipment.

6.1 Alternative Choices (Solutions) to Dilemmas and Complexities

Solutions to challenges include but not limited to: selecting and procuring appropriate equipment and tools for the job; supporting capacity building through training; recruitment and/or inclusion of engineering professionals in management teams; deployment of automation where feasible; provision of personal protective equipment; participating in lobbying and formulation of policies; and allocation of adequate financial resources.

7. WORKING CONDITIONS/ENVIRONMENT

Wood Processing Technologists work with a variety of machinery, toxic substances, and volatile materials. Their work environment is susceptible to fires, explosions, structural failures, and equipment malfunctions. Working conditions include: cold, hot, and wet conditions, climbing heights, standing and/or walking for long hours, lifting materials, working in day or night shifts, and working in areas that are noisy and dusty, as well as areas with limited lighting and ventilation, etc.

8. PARTIES INVOLVED/INTERACTING WITH THE JOBHOLDER OR TRAINEE

8.1 Internal/Within the Organisation

Management, supervisors, subordinates, and other section members, etc.

8.2 External/Outside the Organisation

Government regulators, professional bodies, clients and suppliers, fellow Wood Processing Technologists from other companies, labour unions, students and interns, etc.

9. PHYSICAL DEMANDS ON THE BODY

- Physique to sustain strenuous conditions;
- Be able to walk and stand for long periods of time;
- Bend, stretch, twist, or reach out;
- Be able to lift relatively heavy woods, tools, and equipment;
- Be able to use fingers, hands, and feet with ease to complete the assigned task
- (dexterity);
- Etc.

ANNEX A

Criteria for Assessments based on this NOS

A.1 Guidelines for Assessment

A.1.1 Criteria for assessment for curricula and learning programmes based on this NOS will be created by curricula and programme developers. Each Performance Criteria (PC) will be assigned marks proportional to its importance in the NOS. Curricula and programme developers will also lay down proportion of marks for theory and practical skills for each performance criteria, giving more weight to practical skills.

There shall be allocated the 'Total Mark', which will be the sum of all marks in each Unit, distributed across the number of PCs in that particular unit. The 'Out of' mark will be the mark allocated to each PC, which will be shared between theory and skills practical assessments.

A.1.2 Individual awarding/assessment bodies or institutions and other users of the NOS will create unique question papers for the theory part, and evaluations for skill practical part for their respective candidates.

ANNEX B

NOS Version Control

This Annex gives details necessary for the tracking of the NOS versions based on the number of revisions.

NOS Code	NOS.WPT.01		
ZQF Level	6	Version Number	01
Sector	Manufacturing	Date of Approval	February 2024
Sub Sector	Wood Product Manufacturing	Date of Last Review	N/A
Occupation	Wood Processing Technologist	Date of Next Review	February 2029



*“You learn,
We Standardise”*

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